

Name: _____

Date: _____

Standards: MA.5.AR.2.1-2.4

Show your work in the box for each problem.

Write your final answer on the answer line (or in the blanks shown).

Use the words and strategies from Unit 02: writing and evaluating expressions.

1 Evaluate with Parentheses

Evaluate each expression.

$$(9 + 3) \times 4 = \underline{\hspace{2cm}} \quad | \quad 6 \times (15 - 8) = \underline{\hspace{2cm}} \quad | \quad (7 + 5) \div 4 = \underline{\hspace{2cm}}$$

Answers _____

2 Order of Operations

Evaluate step by step.

$$5 + 3 \times 8 - 4 = \underline{\hspace{2cm}} \quad | \quad 36 \div 6 + 2 \times 5 = \underline{\hspace{2cm}}$$

Answers _____

3 With Exponents

Evaluate.

$$4^2 + (7 - 3) \times 5 = \underline{\hspace{2cm}} \quad | \quad (2 + 3)^2 - 8 = \underline{\hspace{2cm}}$$

Answers _____

4 Write the Expression

Write an expression (do not evaluate) for each phrase.

(A) Four times the difference of 18 and 6

(B) The square of the sum of 3 and 2, decreased by 9

SHOW YOUR WORK

Expressions _____

5 Bake Sale

A baker makes 5 trays of 12 cupcakes and 3 trays of 8 muffins. She sells 25 items.

Write one expression for the number of items remaining. Evaluate it.

Expression and Answer _____

6 Comparing Expressions

Without computing, predict which is greater:

$5 \times (8 + 2)$ vs. $5 \times 8 + 2$. Then calculate to verify.

Prediction and Verification _____

7 Find the Mistake

Atlas evaluates $3 + 4^2 \times 2$ as follows: $3 + 4 = 7$, $7^2 = 49$, $49 \times 2 = 98$.

Find Atlas's error. Show the correct step-by-step solution.

Correct Solution _____

8 Insert Parentheses

Add parentheses to make each equation true.

$$2 + 6 \times 3 + 1 = 32 \quad | \quad 10 - 4 \times 2 = 12 \quad | \quad 3 \times 4 + 5 \times 2 = 54$$

Answers _____

9 Multi-Step Word Problem

Mira earns \$12 per hour. She works 6 hours on Saturday and 4 hours on Sunday. She spends \$35 on supplies.

Write one expression and evaluate it to find how much money she has left.

Expression and Answer _____

10 Same Value?

Kai claims: $(4 + 3)^2 = 4^2 + 3^2 = 16 + 9 = 25$.

Is Kai correct? Evaluate $(4 + 3)^2$ correctly and explain why the result differs from $4^2 + 3^2$.

Explanation _____